Samuele Pironaci

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PROFILE

Computer Science and AI student at the University of Calabria. I mostly build backends in Java (Spring Boot) and Node.js on top of PostgreSQL and MongoDB, and I also write C++. Lately I have been working on mobile too, with Kotlin Multiplatform. I like designing clean REST APIs and then building the interface that uses them.

EDUCATION

University of Calabria In progress

Bachelor's degree in Computer Science and Artificial Intelligence

Relevant courses: Data Structures and Algorithms, Discrete Mathematics, Databases, Operating Systems, Web Applications, User Interface Design, Networks and Security, Parallel Algorithms and Distributed Systems, Artificial Intelligence.

PROJECTS

GeoSentinel Sep 2026 - present

Personal project · mobile app and backend

Java 21, Spring Boot 4, PostgreSQL + PostGIS, Redis, MinIO · Kotlin Multiplatform, Compose Multiplatform

- Trail safety platform for mountain parks. Hikers see trails on a map, report hazards with photos, book mountain huts and send an SOS; park managers and rescue teams review reports, handle maintenance tickets and dispatch squads.
- Backend in Spring Boot with a pragmatic hexagonal architecture. Spatial queries on trails and hazards run on PostGIS through Hibernate Spatial; schema changes go through Flyway migrations.
- Security: JWT authentication as an OAuth2 resource server, opaque refresh tokens with rotation, rate limiting in Redis with Lua scripts, idempotency keys on SOS and bookings, errors in RFC 7807 format.
- Photos upload straight to MinIO with presigned POST policies. Push notifications go through Firebase Cloud Messaging. The OpenAPI contract (springdoc) is the source the app generates its API client from.
- Android and iOS app sharing one Kotlin codebase: Ktor for networking, Koin for DI, MapLibre maps, a Room cache with ETags, and an offline outbox that sends bookings and SOS once the connection comes back. Tokens are kept in Android Keystore and iOS Keychain.
- Integration tests with Testcontainers, 300+ tests on the app, CI on GitHub Actions. Production runs with Docker Compose behind Caddy (automatic HTTPS), with daily backups and OpenTelemetry tracing. Design choices are recorded in ADRs and arc42/C4 docs.

Ricette Facili (recipe blog) Jan 2026

Spring Boot, Angular, PostgreSQL

- Web app with a Spring Boot REST backend and a separate Angular frontend, with data stored in PostgreSQL.
- Designed the REST API and built a responsive interface on top of it.

Food Delivery+ Sep 2025

Java, JDBC, SQL

- Java desktop application structured with MVC, plus the DAO and Singleton patterns, keeping business logic separate from the UI and the data layer.
- Data access with JDBC and plain SQL, with attention to OOP principles and code that stays easy to maintain.

Multi-agent AI: Tron edition

Answer Set Programming, DLV2, EmbASP, MVC

- Real-time game where four AI agents compete on a grid. The game state lives in an MVC model, while each agent's decisions come from a separate declarative logic layer.
- Agent logic written in ASP and solved by DLV2 through EmbASP. Each tick the solver weighs thousands of possible moves to avoid borders, trails and other agents; integrity constraints rule out U-turns and head-on collisions.

TECHNICAL SKILLS

Languages	Java, Kotlin, C/C++, JavaScript, TypeScript, Python, SQL, ASP (Answer Set Programming), HTML, CSS, Rust (learning)
Backend	Spring Boot (Web, Security, Data JPA), Hibernate, Flyway, MapStruct, Node.js, Express, REST, OpenAPI, JWT / OAuth2
Mobile & frontend	Kotlin Multiplatform, Compose Multiplatform, Ktor, Koin, Room, React, Angular
Databases & storage	PostgreSQL, PostGIS, MySQL, MongoDB, Redis, MinIO (S3)
DevOps & cloud	Docker, Docker Compose, GitHub Actions (CI/CD), Caddy, Firebase Cloud Messaging, OpenTelemetry
Testing	JUnit, Testcontainers, Robolectric
Parallel computing	MPI, POSIX Threads
Tools	Git, GitHub, IntelliJ IDEA

WORK EXPERIENCE

Waiter and cashier, Chocolat, Soverato (part-time) Jun - Aug 2023

Waiter, Drop, Soverato (part-time) Jun - Aug 2022

SOFT SKILLS

Practical problem solving, picking up new tools quickly, open to feedback and code review, clear communication in a team.